

EXECUTIVE TECHNOLOGY
PORTFOLIO

Mahmoud Salama

Senior Technology Executive | Digital Transformation |
Enterprise Systems | Software & Architecture

*National platforms. Digital government. Enterprise delivery.
Operational ownership.*

SELECTED, VERIFIED PROJECT HISTORY

Egypt & Oman | 2008–Present



18+

years

Extensive

Keyframe projects

54

technology team

National scale

monthly transactions

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EXECUTIVE PROFILE

Technology leadership at national scale

A hands-on technology executive who has progressed from Linux and full-stack engineering to enterprise architecture, digital transformation and end-to-end technology leadership across Egypt and Oman.



Leadership proposition

Strategy translated into delivery

Connects operating priorities, architecture, roadmaps, requirements, engineering, QA, release, adoption and service improvement.

Platforms, not isolated projects

Builds reusable, configurable capabilities and shared services that support multiple agencies, workflows, channels and operating teams.

Executive scope with technical depth

Retains direct fluency in application architecture, databases, APIs, workflows, integrations, reporting and production problem-solving.

Operational accountability

Owns production readiness, support, customer service, training, incident escalation, continuity requirements and continuous improvement.

Portfolio reading note

This is a representative portfolio—not an exhaustive catalogue of all work. Employers are separated from government and commercial client assignments, delivery ownership is stated explicitly, and confidential or undocumented details are intentionally excluded.

CAREER JOURNEY

From engineer to enterprise technology executive

A career built through progressively wider ownership: software, teams, government platforms, enterprise services, digital transformation and national operations.

2025–Present	UPA Head of the Central Administration for Information Systems & Digital Transformation Leads a 54-person organization across applications, analysis/QA/reporting, support/customer service, architecture, data and transformation governance.
2020–2025	UPA General Manager — Systems, Applications & Technical Support Directed application, platform, support and service delivery; digitized workflows and strengthened governance, release readiness, training and nationwide adoption.
2016–2020	Integral Solutions LLC — Oman Senior Full-Stack Developer — Government Solutions Led delivery teams while remaining hands-on across national education, correspondence, recruitment, incident, agriculture and operational systems.
2011–2016	Keyframe Egypt Development Team Leader & Senior Full-Stack Developer Delivered an extensive portfolio of digital projects and reusable CMS and conference platforms, with hands-on technical ownership across major assignments.
2008–2010	Egyptian Armed Forces Linux Systems & Web Developer Contributed to a custom Linux-from-Scratch distribution and Arabic localization; confidential identifiers are omitted.

Government and client exposure

Current employment is with UPA. Earlier Omani ministries and agencies were Integral Solutions client assignments; Egyptian ministries, authorities and commercial organizations were Keyframe client assignments. This distinction is maintained throughout the portfolio.

Egypt

UPA; Ministries of Health and Planning; MSAD; National Management Institute; Cairo Traffic; healthcare, education, events and commercial clients.

Oman

On-site and delivery assignments across the Ministries of Education, Defence and Agriculture, plus Omani government entities and Al Wahat Club.

Enterprise technology leadership for healthcare supply

One executive scope spanning a unified national procurement platform and a wider portfolio of integrated health, CRM, publishing, reporting, data, integration, support and AI initiatives.



Unified National Procurement Platform

A unified enterprise platform supporting demand consolidation, approvals, tendering, supplier evaluation, award and contracting, purchase orders, delivery and receipt, reporting, auditability, support and shared enterprise services.

- Service and technology leadership across strategy, architecture, delivery governance, operations, integration, data and service performance.
- Multi-level approvals, committees, evaluation, contracts, routing, escalation, SLA timers, maker-checker controls, audit history and role/data-level permissions.
- Configurable workflows and digital records replacing large volumes of paper-, Excel- and email-based processes.
- Cross-functional delivery spanning analysts, architects, engineers, QA/reporting, data, support/customer service and third parties.

Wider enterprise portfolio

Integrated healthcare

HIS and electronic pharmacy capabilities, ICD-10/11 support and integration with multiple health-sector organizations.

Data and integration

Database design, master/reference data, data quality, analytics, REST/SOAP services, data/file exchange and enterprise integration.

Enterprise service channels

CRM, SMS notifications, public websites, CMS/electronic publishing, documents, search, reporting and user communications.

AI and modernization

AI-assisted engineering and enterprise AI initiatives governed as part of the wider technology roadmap.

Digital health, shared services and national operations

Integrated HIS & Electronic Pharmacy Platform

Operational across the health-sector environment | Led analysis, database design and delivery; contributed directly to programming

Reusable healthcare platform for health entities with a complete electronic pharmacy and HIS capabilities, ICD-10 and ICD-11 support and integrations with multiple health-sector organizations.

National Technical Support & Customer-Service Operation

Operational | Service and technology leadership

Approximately 20-person support and customer-service operation handling around 2,000 calls per month, covering technical, functional, data, access, integration and operational incidents, plus onboarding and training nationwide.

Dynamics 365 Integration & Enterprise Data Alignment

Operational | Led the integration workstream through the Central Administration

Unified integration work connecting the authority's platforms with Microsoft Dynamics 365 and related enterprise data and workflow requirements.

Omnichannel SMS Notification Service

Operational across all required channels | Led analysis, delivery and integration

Central notification capability integrated with the authority's platform and service channels.

Electronic Signature Enablement

Implementation in progress; production signing not yet activated | Led requirements, workflows, integration readiness and governance

Prepared platform workflows and integration components for electronic signature while keeping the production status explicit.

Enterprise Data, Reporting & Analytics Environment

Operational | Service and technology leadership and hands-on technical contributor

Database, reporting, data-quality, master-data, integration and analytics capabilities supporting operational and executive decisions across large national datasets.

Service model

A support and customer-service function of approximately 20 people—within the fixed 54-person organization—handles around 2,000 calls each month across technical, functional, data, access, integration and operational issues, alongside onboarding and training.

UPA OPERATING MODEL

Governance, availability and delivery boundaries

Executive accountability is expressed precisely: application and platform ownership remains internal, while infrastructure services are consumed from an independent data-centre provider.

Internally governed

Applications, platform configurations, business analysis, architecture, engineering, QA/UAT, data/reporting, integrations, service operations, support, training and selected delivery settings.

Application delivery controls

Release readiness, security requirements, access controls, auditability, monitoring/visibility requirements, incident/RCA practices, support procedures and change governance.

Data-centre service boundary

Operating systems, networks and firewalls are operated by an independent, geographically separate data-centre service. Infrastructure availability itself is not claimed as an internal operations function.

Continuity and recovery

Periodic disaster-recovery tests are performed under Mahmoud's supervision. Geographic DR is available; failover remains manually controlled.

Leadership operating span

- Technology strategy, operating-model design and program/portfolio prioritization.
- Enterprise and solution architecture, application modernization and integration governance.
- Requirements, process redesign, software delivery, QA/UAT, reporting and production readiness.
- Customer service, technical support, CRM case handling, training, adoption and escalation.
- Data governance, reference data, reporting architecture, operational analytics and executive evidence.
- Stakeholder alignment across government entities, healthcare organizations, vendors and third parties.

Documented leadership appointments

- Digital Transformation Committee Member - World Bank Group B-READY Egypt Assessment (2025-2026)
- Board Member - National Blood Operations Oversight (2021-2024)
- Governance Lead - National Pharmaceutical Track & Trace Program (2021-2024)

Government delivery in Oman

Senior Full-Stack Developer — Government Solutions. Mahmoud remained a hands-on technical contributor across government client assignments throughout the period.

Project / Platform	Context	Ownership	Evidence-based scope
Oman National Educational Portal	Oman Ministry of Education	Core technical contributor	National education platform serving approximately one million users; contributed across analysis, application and database development, reporting, testing, deployment, rollout and support.
Maktabi Electronic Correspondence Management	Omani government entities	Core technical contributor	Large electronic correspondence platform covering creation, routing, forwarding, tracking, management, reporting, supporting database components, testing, production deployment, release implementation and support.
Recruitment System	Oman Ministry of Education	Technical owner	Built the complete recruitment platform independently from analysis and design through database, development, testing and production deployment; recipient of the Sultan Qaboos Award.
Student Incident Management System	Oman Ministry of Education	Technical owner	End-to-end system for recording, following up and reporting student incidents across the education environment.
International Schools Management	Oman Ministry of Education	Core technical contributor	Contributed across requirements, application and database development, testing, deployment and support.
IT Asset & Hardware Management	Oman Ministry of Education	Core technical contributor	Asset-management solution focused on technology hardware, inventory and operational control.
Ministry of Agriculture Portal & Omani Dates Market	Oman Ministry of Agriculture	Technical owner	Designed and built the portal and dates-market capabilities, including purchasing and request workflows without an online payment gateway.
Al Wahat Club Management System	Royal Guard of Oman club assignment	Core technical contributor	New system covering memberships and affiliates, hall and facility reservations, approval workflows and event management. This was a client project, not employment at the club.

Delivery approach

- Led development teams while remaining hands-on across analysis, databases, C#/.NET development, reporting, testing, deployment, rollout and support; worked substantially on-site across Omani government assignments.
- Delivered the Recruitment System and Student Incident Management System independently; Integral Solutions was the employer and the ministries and agencies were client assignments.

Technology used

C#, .NET / ASP.NET Web Forms, Web API, Entity Framework, LINQ, SQL Server, JavaScript, jQuery, AJAX, Kendo UI, Telerik UI and Reporting, Infragistics, DevExpress, Crystal Reports, REST/SOAP and IIS.

Extensive project delivery and reusable product foundations

Development Team Leader & Senior Full-Stack Developer. Sole end-to-end ownership for the confirmed projects presented here, plus leadership of a seven-person full-time team during the final three years. During this period he also founded the Bug Systems concept and independently delivered the bilingual Salamat Healthcare Platform; it is not Keyframe client employment.

Extensive

digital projects

7

full-time team members

2

ITIDA award-winning solutions

4

Electrolux country sites

Reusable .NET CMS & electronic publishing

Designed and built a reusable CMS supporting multilingual and RTL publishing, pages, articles, news, events, photo/video galleries, permissions, workflows, search, SEO, analytics, resource files, reporting and integration extensions. The platform was adapted across government, healthcare, education, events and multi-country commercial portals.

Reusable conference management

Built a reusable platform for Arab Medical Union, UCHID, BGICC and Arab Organizers, covering registration, attendance, schedules, sessions, speakers, research evaluation, certificates, communications, administration and reporting. No payment-gateway claim is made.

Representative individual deliveries

Egypt Ministry of Health & Population Portal

Large government health portal with CMS, services, news and events, electronic library, FAQ, newsletter, polls, media galleries, social integration, reporting, security and deployment.

Egypt Government Digital Portals

Representative solutions include MSAD, Rate Your Services, National Management Institute, the National Coordinating Committee for Combating Corruption, Egypt Calendar, Elections in Egypt and online survey systems. These were client projects, not government employment.

Traffic Reporting & Officers' Affairs System

Complete system for traffic reports and officers' affairs, delivered independently from analysis and database design through testing and deployment.

Egypt Ministry of Planning Portal

Independently handled analysis, SRS, architecture, database, CMS, development, testing and deployment. The specification covered services, news, media, newsletter, tenders, polls, careers, accessibility, security, performance and backup/restore.

Yashfeen Medical Portal

Full medical portal built independently for Keyframe. It achieved an award-recognized solution in an ITIDA competition. Mahmoud delivered two ITIDA award-winning solutions during the Keyframe period; the second project name is not currently documented.

Electronic Examinations & Educational Review System

Complete electronic-examination and curriculum-review system for multiple educational certificate curricula.

SELECTED PROJECT DIRECTORY

Government, health, education, events and enterprise platforms

Representative projects. Smaller projects from the broader Keyframe portfolio are intentionally omitted.

Project / Platform	Context	Ownership	Evidence-based scope
Unified National Procurement Platform	UPA	Led through the Central Administration	Unified national platform covering demand consolidation, approvals, tendering, supplier evaluation, award and contracting, purchase orders, delivery and receipt, reporting, auditability, support and shared enterprise services.
Integrated HIS & Electronic Pharmacy Platform	UPA	Led through the Central Administration	Reusable healthcare platform for health entities with a complete electronic pharmacy and HIS capabilities, ICD-10 and ICD-11 support and integrations with multiple health-sector organizations.
Dynamics 365 Integration & Enterprise Data Alignment	UPA	Led through the Central Administration	Unified integration work connecting the authority's platforms with Microsoft Dynamics 365 and related enterprise data and workflow requirements.
Omnichannel SMS Notification Service	UPA	Led through the Central Administration	Central notification capability integrated with the authority's platform and service channels.
Electronic Signature Enablement	UPA	Led through the Central Administration	Prepared platform workflows and integration components for electronic signature while keeping the production status explicit.
Oman National Educational Portal	Oman Ministry of Education	Core technical contributor	National education platform serving approximately one million users; contributed across analysis, application and database development, reporting, testing, deployment, rollout and support.
Maktabi Electronic Correspondence Management	Omani government entities	Core technical contributor	Large electronic correspondence platform covering creation, routing, forwarding, tracking, management, reporting, supporting database components, testing, production deployment, release implementation and support.
Recruitment System	Oman Ministry of Education	Technical owner	Built the complete recruitment platform independently from analysis and design through database, development, testing and production deployment; recipient of the Sultan Qaboos Award.
Student Incident Management System	Oman Ministry of Education	Technical owner	End-to-end system for recording, following up and reporting student incidents across the education environment.
International Schools Management	Oman Ministry of Education	Core technical contributor	Contributed across requirements, application and database development, testing, deployment and support.
IT Asset & Hardware Management	Oman Ministry of Education	Core technical contributor	Asset-management solution focused on technology hardware, inventory and operational control.
Ministry of Agriculture Portal & Omani Dates Market	Oman Ministry of Agriculture	Technical owner	Designed and built the portal and dates-market capabilities, including purchasing and request workflows without an online payment gateway.

SELECTED PROJECT DIRECTORY — CONTINUED

Project / Platform	Context	Ownership	Evidence-based scope
Al Wahat Club Management System	Royal Guard of Oman club assignment	Core technical contributor	New system covering memberships and affiliates, hall and facility reservations, approval workflows and event management. This was a client project, not employment at the club.
Reusable .NET CMS & Electronic-Publishing Platform	Keyframe Egypt	Technical owner	Custom reusable CMS supporting multilingual and RTL publishing, pages, articles, news, events, photo and video galleries, permissions, workflows, search, SEO, analytics, resource files and integration extensions.
Egypt Ministry of Health & Population Portal	Egypt Ministry of Health & Population	Technical owner	Large government health portal with CMS, services, news and events, electronic library, FAQ, newsletter, polls, media galleries, social integration, reporting, security and deployment.
Egypt Ministry of Planning Portal	Egypt Ministry of Planning	Technical owner	Independently handled analysis, SRS, architecture, database, CMS, development, testing and deployment. The specification covered services, news, media, newsletter, tenders, polls, careers, accessibility, security, performance and backup/restore.
Egypt Government Digital Portals	Keyframe Egypt	Technical owner	Representative solutions include MSAD, Rate Your Services, National Management Institute, the National Coordinating Committee for Combating Corruption, Egypt Calendar, Elections in Egypt and online survey systems. These were client projects, not government employment.
Reusable Conference Management Platform	Keyframe Egypt	Technical owner	Conference platform covering registration, attendance, schedules, sessions, speakers, research evaluation, certificates, communications, administration and reporting. No payment-gateway claim is made.
Yashfeen Medical Portal	Keyframe product	Technical owner	Full medical portal built independently for Keyframe. It achieved a award-recognized solution in an ITIDA competition. Mahmoud delivered two ITIDA award-winning solutions during the Keyframe period; the second project name is not currently documented.
Traffic Reporting & Officers' Affairs System	Cairo Traffic Headquarters	Technical owner	Complete system for traffic reports and officers' affairs, delivered independently from analysis and database design through testing and deployment.
Electronic Examinations & Educational Review System	Keyframe Egypt	Technical owner	Complete electronic-examination and curriculum-review system for multiple educational certificate curricula.
Electrolux Multi-Country Websites	Electrolux	Technical owner	Reusable CMS implementation customized for Egypt, Saudi Arabia, the UAE and Turkey.
Salamat Healthcare Platform	Bug Systems concept	Technical owner	Bilingual healthcare platform developed as part of a virtual private-company concept. It is presented separately from Keyframe client employment.

Directory interpretation

Sector coverage

Government, healthcare, education, events, electronic publishing and multi-country commercial delivery.

Technology range

C#/.NET, SQL platforms, JavaScript, APIs, workflows, CMS/DMS, reporting, integration and production delivery.

Ownership model

Current executive ownership, earlier team delivery and independently owned end-to-end projects are stated separately.

Evidence boundary

This directory contains documented representative work; smaller projects within the broader Keyframe portfolio remain omitted.

CASE STUDIES

National platforms and award-winning delivery

CASE STUDY 01 Unified National Procurement Platform

CONTEXT	UPA Egypt Operational national platform
CHALLENGE	Coordinate a complex procurement lifecycle across a large user and entity base while replacing fragmented manual, paper, spreadsheet and email processes.
RESULT	
CAPABILITIES	• Led platform strategy, architecture, delivery governance, data, integration, service readiness and operational ownership through the Central Administration. • Aligned analysts, engineers, QA/reporting, support/customer service, stakeholders and third parties around configurable workflows and shared controls. • Combined executive oversight with hands-on escalation across requirements, data, integration, production behavior and service improvement. Operational platform serving approximately 60,000 users, 90,000 facilities/entities and 300 suppliers, processing around one million transactions per month. Digital transformation • Enterprise architecture • Platforms • Data • Integration • Service operations

CASE STUDY 02 Integrated HIS & Electronic Pharmacy

CONTEXT	UPA Digital health Operational across the health-sector environment
CHALLENGE	Create a reusable healthcare capability that can operate in different health entities and integrate with multiple health-sector organizations.
RESULT	
CAPABILITIES	• Participated from analysis through delivery; designed the database, led major workstreams and contributed directly to programming. • Supported complete electronic-pharmacy and HIS capabilities, including ICD-10 and ICD-11. • Kept scope accurate: inpatient admission is not claimed. A reusable operational healthcare platform active across the health-sector environment. Digital health • Database architecture • Software delivery • Integration • Clinical coding

CASE STUDIES

Government platforms in Oman and Egypt

CASE STUDY 03 Oman National Educational Portal

CONTEXT

Integral Solutions | Oman Ministry of Education | Core technical contributor

CHALLENGE

Deliver and operate a national education platform at large user scale across complex government requirements and rollout needs.

RESULT

CAPABILITIES

- Led development teams throughout the Integral period while remaining a hands-on senior developer.
- Contributed across analysis, application and database development, reporting, testing, deployment, support and rollout.
- Worked within the employer/client boundary: Integral Solutions was the employer; the Ministry was the client assignment.

Operational national portal serving approximately one million users.

Government platforms • Team leadership • C#/.NET • SQL Server • Delivery • Support

CASE STUDY 04 Recruitment System

CONTEXT

Integral Solutions | Oman Ministry of Education | Technical owner

CHALLENGE

Build a complete government recruitment solution independently, covering the full journey from requirements to production.

RESULT

CAPABILITIES

- Owned analysis, design, architecture, database, development, testing and deployment as the sole end-to-end developer.
- Delivered on the established Integral government stack and operational environment.
- Supported formal government evaluation and production use.

Operational recruitment platform and recipient of the Sultan Qaboos Award.

End-to-end ownership • Government HR • C#/.NET • SQL Server • Award-winning delivery

CASE STUDIES

Reusable platforms and health innovation

CASE STUDY 05 Reusable .NET CMS & Publishing Platform

CONTEXT

Keyframe Egypt | Reused across a broad project portfolio

CHALLENGE

Serve many organizations and sectors without rebuilding common publishing, workflow, multilingual and administration capabilities each time.

RESULT
CAPABILITIES

- Designed and developed the CMS foundation independently.
- Built configurable multilingual/RTL content, media, events, permissions, workflows, search, SEO, analytics and integration components.
- Adapted the foundation to ministries, healthcare, education, events and Electrolux sites in Egypt, Saudi Arabia, the UAE and Turkey.

A reusable product foundation supporting rapid delivery across a broad portfolio of digital projects.

Product architecture • Reuse • CMS • Multilingual/RTL • Government and commercial delivery

CASE STUDY 06 Yashfeen Medical Portal

CONTEXT

Keyframe Egypt | Technical owner | Digital health

CHALLENGE

Create a comprehensive medical portal as a Keyframe product and take it through external technology evaluation.

RESULT
CAPABILITIES

- Owned the complete delivery independently—from analysis and data design through implementation, testing and deployment.
- Applied the reusable CMS/platform approach to a full healthcare experience.
- Delivered an ITIDA award-recognized solution during the Keyframe period.

Achieved a award-recognized solution in an ITIDA competition. The second award-winning solution is referenced without an undocumented project name.

Digital health • Technical owner • Product engineering • ITIDA award

CAPABILITIES & RECOGNITION

A practical executive–engineering bridge

Strategy & governance

Technology strategy, digital transformation, roadmaps, portfolio prioritization, operating models, executive stakeholders, risk, controls and NFRs.

Engineering & delivery

C#, .NET, ASP.NET Web Forms/Core/Web API, JavaScript/React, SQL Server/PostgreSQL/MySQL, SDLC, QA/UAT, CI/CD, release and go-live.

Service operations

Technical support, customer service, service desk/call center, incident/RCA, change/release, training, adoption and business continuity/DR oversight.

Architecture

Enterprise and solution architecture, modular/multi-tenant/configurable platforms, API-first design, workflow/RBAC, integration and data architecture.

Data & integration

Database design, master/reference data, data quality, reporting architecture, REST/SOAP, JSON, API contracts and data/file integrations.

AI-enabled work

Generative AI, prompt engineering and AI-assisted coding, analysis, documentation, SQL, prototyping and enterprise AI adoption strategy.

Awards and governance appointments

- Two ITIDA award-winning solutions during the Keyframe period, including Yashfeen with a award-recognized solution
- Sultan Qaboos Award received by the Oman Ministry of Education Recruitment System
- Digital Transformation Committee Member - World Bank Group B-READY Egypt Assessment (2025-2026)
- Board Member - National Blood Operations Oversight (2021-2024)
- Governance Lead - National Pharmaceutical Track & Trace Program (2021-2024)

Education & languages

B.Sc. in Computer Science, Faculty of Computers & Informatics, Zagazig University | 2003–2007

Arabic: Native | English: Professional working proficiency

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